Discoveries
Designed for curious minds

DRIFT CIPHER CHALLENGE

Decode the hidden words inspired by your racing notebook cover. Each clue points to something you can spot on the cover—but the words are locked behind a shifting code.

How the code works:
Each number = a letter (A=1, B=2 ... Z=26).
BUT the message is shifted forward by its position:
1st letter shifted +1, 2nd +2, 3rd +3 ...
You must shift backward to reveal the real answer.

...makes when tyres scream but control stays ...r

_____

...tern of victory that flutters in squares of ...

_____

...u alone—the phrase on the cover says ...

_____

...ng everything forward in a race—feel it ...

_____

5. Clue: The wild white trails behind the tyres—clouds of power on the cover.
Code: 20 15 18 15 10
Answer: _____

6. Clue: The machine at the heart of the cover—loud, glossy, and ready to bolt.
Code: 19 3 6 9 4 3 21
Answer: _____



*Whenever you see "cover," it means both the front and back

7. Clue: The glowing circle behind the car—not a light, not a wheel, but the sky's own badge.
Code: 20 23 17
Answer: _____

8. Clue: Where tyres grip and skid—the racing world beneath the wheels.
Code: 21 20 4 7 16
Answer: _____

THE DRIFTING FACT

...g began as an underground racing technique in Japan—now it's a global ...motorsport where drivers are judged not just on speed, but on style.

...KOMM31125 B

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CELL AND ITS ORGANELLES

classmate
Date
Page 1

Introduction

The cell is the basic structural and functional unit of the human body. The human body is made up of trillions of cells, which are organized into tissues, organs and organ systems. Although cells differ greatly in their size, shape and function, most human cells have a common basic organization.

A typical human cell consists of:

1. PLASMA MEMBRANE
2. CYTOPLASM
3. NUCLEUS

The cytoplasm contains various specialized structure called cell organelles. Each organelle performs a specific function, and together they maintain the normal activities and survival of the cell.

General Structure of a Cell

A typical human cell can be divided into three major components:

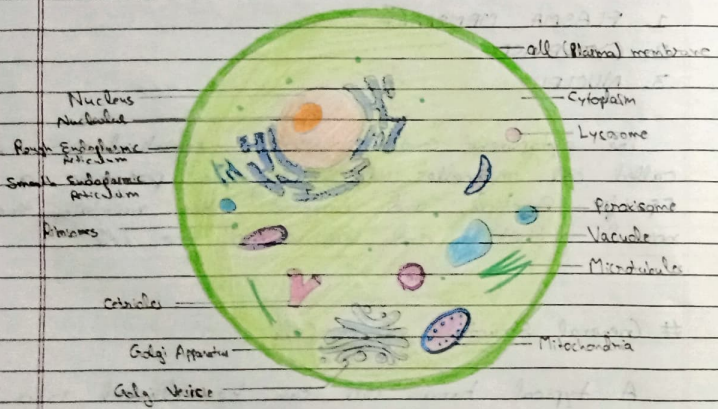
1 Plasma membrane

The plasma membrane forms the outer boundary of the cell. It separates the intracellular environment from the extracellular environment.

It is mainly composed of:

- Lipids
- Proteins
- Carbohydrates

The plasma membrane is selectively permeable and regulates the movement of substances into and out of the cell.



ANIMAL CELL DIAGRAM

2. Cytoplasm

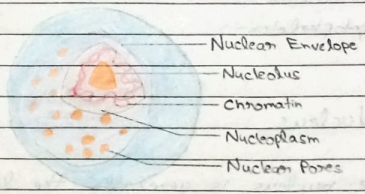
The cytoplasm consists of:

- Cytosol - the fluid component
- Cell organelles
- Cytoplasmic inclusions

The cytoplasm is the site of many important metabolic reactions.

3. Nucleus

The nucleus contains most of the genetic material of the cell in the form of DNA. It controls many cellular activities by regulation of gene expression and protein synthesis



Nucleus

Cell Organelles

Cell organelles are specialized structure present within the cytoplasm and nucleus that perform specific functions

They can broadly be classified as:

Membrane - Bound organelles

- Mitochadeia
- Endoplasmic reticulum
- Golgi apparatus
- Lysosomes
- Peroxisomes
- Nucleus

Non- membrane - bound structures

- Ribosomes
- Centrosome
- Cytoskeleton

Nucleus

The nucleus is generally the largest organelle of cell and acts as major control centre of the cell.

Structure

The nucleus consist of:

1. Nuclear envelope
2. Nucleoplasm
3. Chromatin
4. Nucleolus

- Nuclear envelope

The nuclear envelope consists of two membranes and seperates the nucleus from the cytoplasm.

It contains openings called nuclear pores, through which substances such as RNA and proteins can move between the nucleus and cytoplasm.

- Chromatin

Chromatin is composed mainly of DNA and proteins.

During cell division, chromatin becomes highly condensed and forms chromosomes.

- Nucleolus

The nucleolus is a dense region within the nucleus. It is mainly involved in the formation of ribosomal RNA (rRNA) and assembly of ribosomal subunits.

Functions of nucleus

- Stores genetic information
- Controls gene expression
- Regulates protein synthesis
- Controls cell growth and division
- Participates in transmission of hereditary information

Mitochondria

Mitochondria are commonly called the powerhouse of the cell because they generate most of the cell's ATP through aerobic metabolism.

They are present in large numbers in cells with high energy requirements, such as:

- Cardiac muscle cells
- Skeletal muscle cells
- Liver cells

Structure

A mitochondria has:

1. Outer mitochondrial membrane
2. Intermembrane space
3. Inner mitochondrial membrane
4. Cristae
5. Matrix

The inner mitochondrial membrane forms folds called cristae. These increase the surface area available for oxidative phosphorylation.

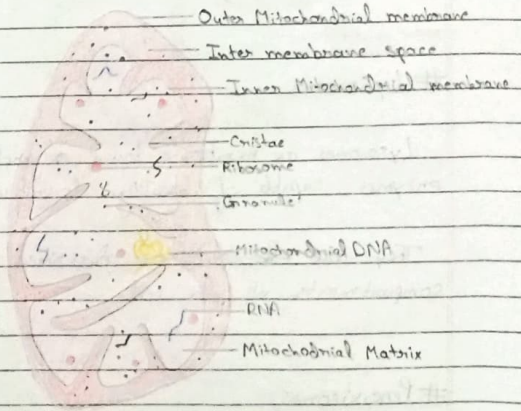
The matrix contains enzyme involved in the citric acid cycle, along with mitochondrial DNA and ribosome

Functions

- ATP production
- Oxidative phosphorylation
- Participation in cellular metabolism
- Regulation of apoptosis
- Some steps of calcium handling

Clinical importance

Mitochondria contains their own DNA. Therefore, disorders affecting mitochondrial function can particularly affect tissue with high energy requirements, especially the brain and muscles.



Mitochondria

Endoplasmic Reticulum

The endoplasmic reticulum (ER) is a network of membrane tubules and sacs present in the cytoplasm.

It is of two main types:

A. Rough Endoplasmic Reticulum
B. Smooth

Golgi Apparatus

The Golgi apparatus is a membrane-bound organelle involved mainly in the modification, sorting and packaging of proteins and lipids.

Lysosomes

Lysosomes are membrane-bound organelles containing hydrolytic enzymes capable of breaking down various substances.

They are sometimes described as the digestive compartments of the cell.

Peroxisomes

Peroxisomes are small membrane-bound organelles containing enzymes involved in several metabolic reaction.

Ribosomes

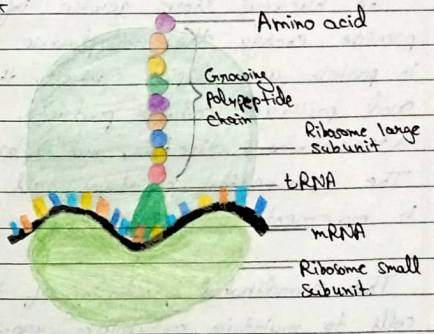
Ribosomes are small, non-membrane-bound structures responsible for protein synthesis.

They consist of:

- Ribosomal RNA (rRNA)
- Proteins

In human cells, cytoplasmic ribosomes are 80S, consisting of:

- 60S large subunits
- 40S small subunit



Ribosome Diagram

Centrosome

The centrosome is an important structure involved in the organization of microtubules.

Cytoskeleton

The cytoskeleton is a network of protein filaments that provides structural support to the cell.

Conclusion

The cell is the basic structural and functional unit of the human body. Its different organelles perform specialized functions that are essential for cellular survival and normal physiological activity.

The nucleus stores genetic information, mitochondria provide energy, the endoplasmic reticulum participates in protein and lipid synthesis, the Golgi apparatus modifies and packages cellular products, lysosome perform intracellular digestion, and ribosomes synthesize proteins. The cytoskeleton maintains cell shape and participates in movement and intracellular transport.

The coordinated functioning of these components enables cells to maintain metabolism, growth, communication, reproduction and homeostasis.

Therefore, understanding the structure and functions of cell and its organelles is fundamental to the study of physiology.

INTERCELLULAR JUNCTIONS,
~~NECROSIS SYSTEM~~
APOPTOSIS

Intercellular Junctions

Introduction

Cells of the human body are organized into tissues and must remain attached to one another and communicate with neighbouring cells. Specialized structures that connect adjacent cells or attach cells to the extracellular matrix are called intercellular junctions.

They provide:

- Mechanical attachment
- Barrier formation
- Cell-to-cell communication
- Maintenance of tissue integrity

Types of Intercellular Junctions

The major types are:

1. Tight junction
2. Adherens junction
3. Desmosome
4. Hemidesmosome
5. Gap junction

1. Tight Junction

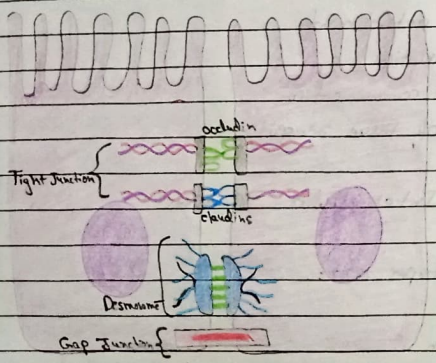
A tight junction, or zonula occludens, forms a seal between adjacent cells, especially in epithelial tissues.

Functions:

- Prevents leakage between cells
- Controls paracellular movement of substances
- Maintains the barrier function of epithelium
- Helps maintain polarity of epithelial cells.

Important proteins include claudins and occludin

Example: Intestinal epithelium



Tight junction in a healthy gut Schematic

2. Adherens Junction

Adherens junctions, or zonula adherens, provide adhesion between neighbouring cells.

They contain cadherins, which are linked to the actin cytoskeleton inside the cell.

Functions:

- Cell-to-cell adhesion
- Maintenance of tissue architecture
- Helps cells withstand mechanical stress
- Participates in changes in cell shape

3. Desmosome

A desmosome, or macula adherens, is a strong spot-like junction between adjacent cells

It is connected to intermediate filaments and provides strong mechanical attachment.

Functions:

- Prevents separation of cells
- Provides mechanical strength
- Important in tissue exposed to stress

Example: Skin and cardiac muscle.

4. Hemidesmosome

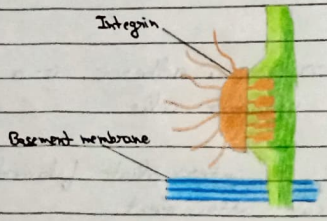
Hemidesmosomes attach epithelial cells to the basement

membrane.

They involve integrins and connect the cell cytoskeleton with the extracellular matrix.

Functions:

- Anchors epithelial cells to the basement membrane
- Provides mechanical stability
- Prevents detachment of epithelial cells.



Hemidesmosome

5. Gap Junction

Gap junction provide direct communication between neighbouring cells

They contain channels formed by proteins called connexin Six connexins form a connexon

Small ions and molecules can pass through these channels

Necrosis

Introduction

Necrosis is a form of cell death caused by severe and irreversible cell injury. It occurs when a cell is damaged beyond the point of recovery.

Unlike apoptosis, necrosis is generally associated with loss of cell membrane integrity and an inflammatory response in the surrounding tissue.

Common causes include:

- Ischaemia and hypoxia
- Physical injury
- Chemical or toxic substances
- Extreme temperatures
- Certain pathological conditions

Morphological Features of Necrosis

The important changes in a necrotic cell include:

1. Cellular changes
2. Nuclear changes

Types of Necrosis

There are mainly five types of necrosis:

1. Coagulative Necrosis
2. Liquefactive Necrosis
3. Caseous Necrosis
4. Fat Necrosis
5. Fibrinoid Necrosis

Gangrene

Gangrene refers to tissue necrosis involving a considerable area of tissue, usually due to impaired blood supply, infection or both.

Dry gangrene

Usually results from prolonged ischemia. The affected tissue becomes dry and shrivelled

Wet gangrene

Occurs when necrotic tissue becomes infected, resulting in tissue liquefaction and inflammation.

Gas gangrene

A severe form of necrotic tissue becomes infected, resulting in tissue liquefaction and inflammation, associated with gas-producing bacteria, particularly clostridium species

Functions:

- Cell-to-Cell communication
- Passage of ions
- Electrical Coupling

Examples: Cardiac muscle and smooth muscle

Nervous System

Introduction

The nervous system is the major system responsible for receiving information, processing it and producing appropriate responses.

It helps the body to:

- Sense changes in the internal and external environment
- Coordinate body functions
- Control movement
- Maintain homeostasis
- Perform higher functions such as learning and memory

Division of Nervous System

The nervous system is broadly divided into:

A. Central Nervous System (CNS)

It consists of:

- Brain
- Spinal cord

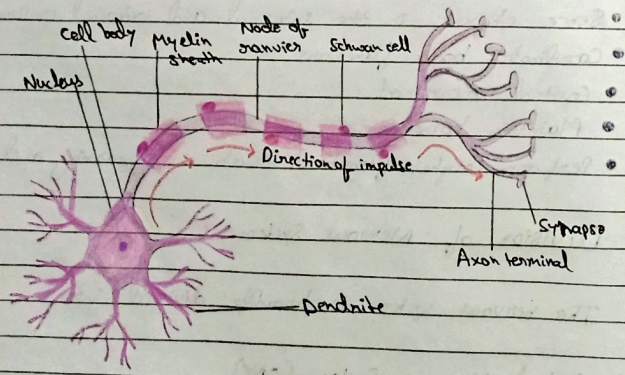
The CNS receives and processes information and coordinates responses.

B. Peripheral Nervous System (PNS)

It consists mainly of:

- Cranial nerves
- Spinal nerves
- Peripheral ganglia

The PNS connects the CNS with different parts of the body.



Neuron

* CNS → Brain + Spinal cord
* PNS → Cranial nerves + Spinal nerves + Ganglia

Cells of the Nervous System

The nervous tissue mainly contains two types of cells:

1. Neurons
2. Neuroglial cells.

1. Neuron

A neuron is the structural and functional unit of the nervous system.

A typical neuron consists of:

1. Cell body
2. Dendrites
3. Axon

Functions

- Receive stimuli
- Generate electrical signals
- Conduct nerve impulses
- Communicate with other neurons, muscle and glands

2. Neuroglia

Neuroglial cells support and protect neurons.

Apoptosis

Introduction

Cells in the human body are continuously being formed and removed. Apoptosis is a regulated form of programmed cell death in which unwanted or damaged cells are eliminated in a controlled manner.

It is an important physiological process in normal development and maintenance of tissue.

Features

1. Cell shrinks
2. Chromatin condenses
3. Nucleus becomes fragmented
4. Cell membrane remains relatively intact
5. Cell breaks into small apoptotic bodies
6. Apoptotic bodies are removed by phagocytic cells

Apoptosis generally occurs without the marked inflammatory response associated with cell injury and necrosis.

Normal cell
↓
Cell shrinkage
↓
Chromatin condensation
↓
Nuclear fragmentation
↓
Apoptotic bodies
↓
Phagocytosis

Importance

Apoptosis is important for:

- Normal embryonic development
- Removal of unwanted cells
- Maintenance of tissue homeostasis
- Removal of damaged cells
- Regulation of immune responses

Example: During development, apoptosis helps separate the digits of the developing limbs.

It also ~~help~~ helps eliminate cells that are no longer required after completion of their function.

Clinical Importance

Abnormal regulation of apoptosis can contribute to disease.

Excessive apoptosis - Excessive loss of cells may contribute to tissue damage and certain degenerative conditions.

Reduced apoptosis - Failure of abnormal cells to undergo apoptosis can contribute to persistence of damaged or abnormal cells and is important in the development of cancer.

Thus, normal regulation of apoptosis is essential for maintaining tissue homeostasis.

Conclusion

- Intercellular junctions, the nervous system and apoptosis are important components of normal human biology.
- Intercellular junctions, the nervous system maintains attachment, barrier function and communication between cells.
- The nervous system receives and processes information and coordinates responses throughout the body. Neurons conduct signals, while neuroglial cells provide support and protection.
- Apoptosis is a controlled form of cell death that removes unwanted or damaged cells and helps maintain normal tissue homeostasis.
- Understanding these basic concepts provides a foundation for the further study of histology, physiology, pathology and other medical subjects.

Introduction

The cells of the human body continuously receive and respond to signals. These signals help cells coordinate their activities, maintain normal body functions and respond to changes in their environment.

Cellular communication is the process by which cells receive, transmit and respond to signals.

Communication may occur:

- Between different cells
- Between a cell and its surrounding environment
- Within the same cell through intracellular signalling pathways

Intracellular signalling converts an external signal into a specific response inside the cell.

1. Types of cellular communication

Depending on the distance between the signalling cell and the target cell, communication can be classified as:

1. Endocrine signalling

Hormones are released into the bloodstream and act on distant target cells.

Example: Insulin released by pancreatic β-cells acts on cells throughout the body.

2. Paracrine Signalling

A cell release signalling molecules that act on nearby cells.

Example: Local chemical mediators acting on neighbouring cells.

3 Autocrine Signalling

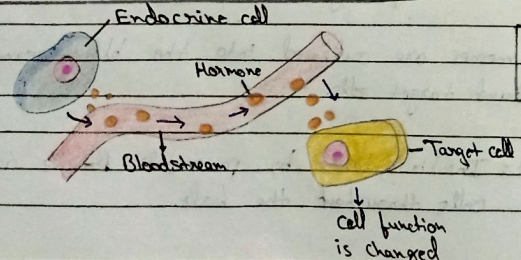
A cell releases a signal that act on the same cell that produced it.

4. Synaptic signalling

A neuron releases a neurotransmitter at a synapse to communicate with another neuron or an effector cell.

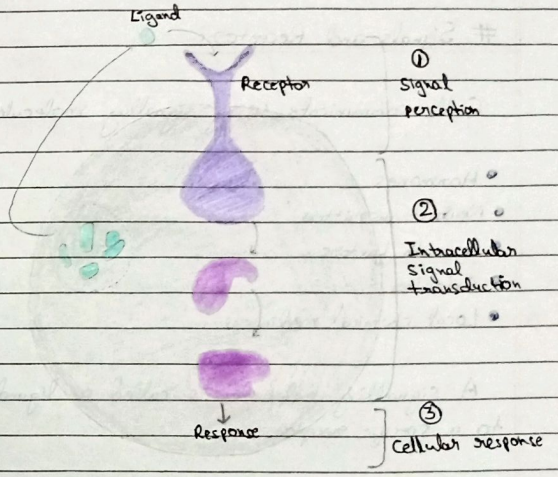
5. Contact-dependent signalling

A signalling molecule present on the membrane of one cell interacts with receptor on an adjacent cell.

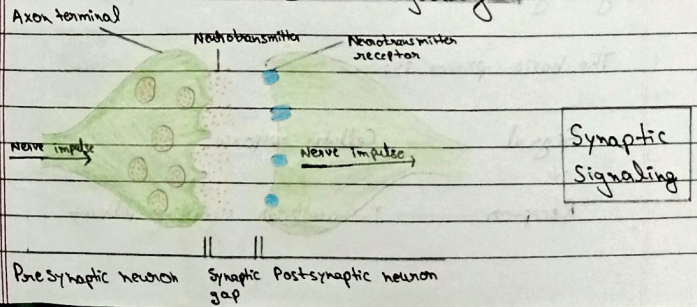


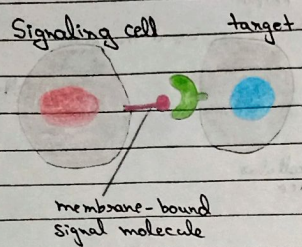


Paracrine Signaling



Autocrine Signaling





Contact Dependent Signaling

Signals and Receptors

Cells communicate using signalling molecules such as:

- Hormones
- Neurotransmitters
- Growth factors
- Cytokines
- Local chemical mediators

A signalling molecule is called a ligand when it binds to a specific receptor.

A receptor is a protein that recognizes a particular signalling molecule and initiates a cellular response.

The basic process is:

Signal → Receptor → Intracellular signalling pathway → Cellular response

Cell Surface Receptors

Some signalling molecules cannot cross the plasma membrane because they are water-soluble. They therefore bind to receptors located on the cell surface.

Important types include:

1. G-protein coupled receptors
2. Enzyme-linked receptors
3. Ion-channel receptors

~~1. G-Protein~~
Second Messengers

A second messenger is an intracellular molecule that helps transmit a signal from a cell-surface receptor to intracellular targets.

Important second messengers include:

- cAMP
- cGMP
- IP_3
- DAG
- Ca^{2+}

They amplify and distribute signals within the cell.

Enzyme - Linked Receptors

Enzyme - linked receptors are membrane receptors that either possess enzymatic activity themselves or are associated with enzymes.

A major example is the receptor tyrosine kinase (RTK)

Mechanism

1. Ligand binds to receptor
2. Receptors come together or undergo conformational change
3. Tyrosine kinase activity is activated
4. Intracellular proteins are phosphorylated
5. Signalling pathways are activated
6. A cellular response occurs.

Example: The insulin receptor is a receptor tyrosine kinase.

RTK signalling is also important for growth factor and cell growth

Intracellular Receptors

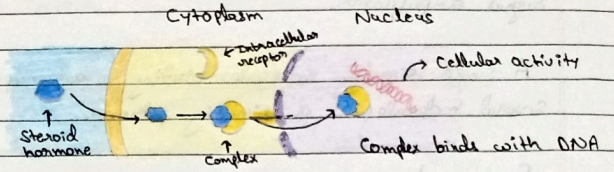
Some signaling molecules are lipid-soluble and can cross the plasma membrane. They bind to receptors located inside the cell.

These receptors may be located in cytoplasm and nucleus of the cell.

Examples of signaling molecules acting through intracellular receptors include:

- Steroid hormones
- Thyroid hormones

The hormone-receptor complex can influence gene expression.



Intracellular receptor binding

Cellular Response

After activation of a receptor, intracellular signaling pathways can produce different responses

These include:

- Changes in enzyme acitivity
- changes in ion movement
- changes in cell metabolism
- changes in gene expression
- cell growth
- Cell division
- Cell differentiation
- Cell movement

Thus, the same basic signaling principle can produce different responses depending on the type of target cell.

Signal Amplification

An important feature of intracellular communication is signal amplification.

A single signaling molecule binding to a receptor can activate several molecules in a signaling pathway.

For example:

One ligand → Receptor activation → Multiple intracellular molecules

↓

large cellular response ← Many second messenger molecules

This allows a relatively small external signal to produce a significant intracellular effect.

Termination of Signal

Cellular signals must eventually be stopped to prevent continuous activation.

Signals may be terminated by:

- Removal or breakdown of the signaling molecule
- Receptor inactivation

- Breakdown of second messengers
- Dephosphorylation of proteins
- Removal of Ca^{2+} from the cytoplasm

For example, cAMP can be broken down by phosphodiesterases.

Termination of signaling is essential for maintaining normal cellular control.

Importance of Intra cellular communication

Intracellular communication is essential for:

- Growth and development: Growth factors regulate cell growth, division and differentiation
- Metabolism: Hormonal signals regulate glucose, lipid and protein metabolism.
- Nervous system: Neurotransmitter allow communication between neurons and between neurons and effector cells.
- Immune response: Cytokines and other mediators regulate immune cell activity.
- Maintenance of homeostasis: Cell signaling allows cells to adjust their activity according to changes in the internal and external environment.

Therefore, understanding intracellular communication is important for understanding both disease mechanisms and pharmacological treatment.

Conclusion

Intracellular communication is an essential mechanism by which cells receive information and convert it into appropriate responses. It involves signaling molecules, receptors, intracellular messengers and signaling pathways.

These mechanisms regulate important processes such as metabolism, growth, secretion, movement, differentiation and maintenance of homeostasis.

A basic understanding of intracellular communication is therefore essential for the study of physiology, biochemistry, pharmacology and pathology.

Introduction

The plasma membrane forms the outer boundary of the cell and separate the intracellular fluid from the extracellular fluid.

It is a selectively permeable membrane, meaning that it allows some substances to cross easily than others.

Transport of substances across the cell membrane is essential for:

- Maintaining the internal environment of the cell
- Obtaining nutrients
- Removing waste products
- Maintaining ionic concentrations.
- Maintaining cell volume
- Cellular communication and normal function.

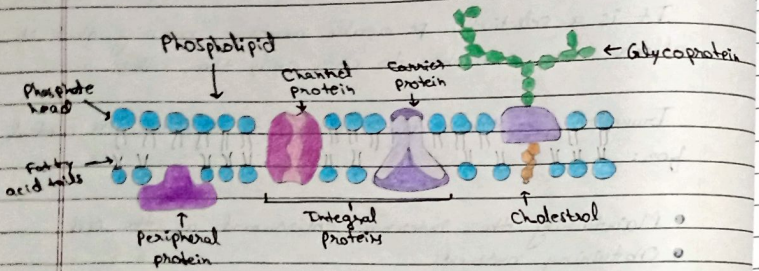
Substances cross the cell membrane through different mechanisms depending on their size, electrical charge, lipid solubility and concentration gradient.

Structure of Plasma membrane

The plasma membrane is mainly composed of:

- Lipids
- Proteins
- Carbohydrates

The basic structure is described by the fluid mosaic model. The lipid bilayer forms the main barriers, while membrane proteins act as channels, carriers, receptors and pumps.



Fluid mosaic model

Types of Membrane Transport

Transport across the cell membrane can broadly be classified into:

1. Passive transport

Does not require direct expenditure of cellular energy. Substances generally move down their concentration or electrochemical gradient.

Examples: A • Simple diffusion
B • Facilitated diffusion
C • Osmosis

2. Active transport

Requires energy and can move substances against their electrochemical gradient

Examples:

A • Primary active transport
B • Secondary active transport

3. Vesicular transport

Large particles or molecules are transported using membrane-bound vesicles.

Examples:

A • Endocytosis
B • Exocytosis

1.A. Simple Diffusion

Simple diffusion is the movement of molecules from an area of higher concentration to an area of lower concentration until equilibrium is approached.

- It does not require ATP or a transport protein.

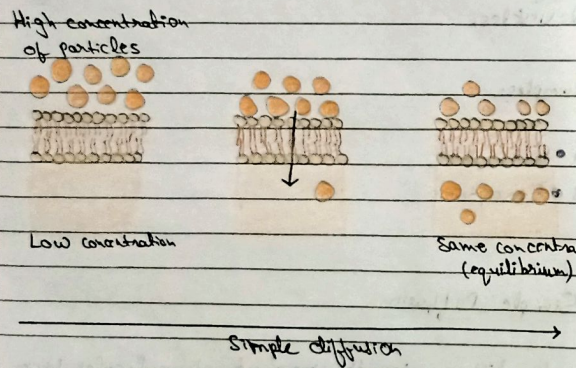
- Small lipid-soluble molecules can easily cross the lipid bilayer.

Examples

- Oxygen
- Carbon dioxide
- Some lipid-soluble substances

The rate of diffusion depends on factors such as:

- Concentration gradient
- Surface area
- Thickness of membrane
- Temperature
- Lipid solubility of the substance



1.b. Facilitated Diffusion

Facilitated diffusion is the movement of a substance down its concentration or electrochemical gradient with the help

of a membrane protein.

- It does not require direct expenditure of ATP.

Membrane proteins involved may function as:

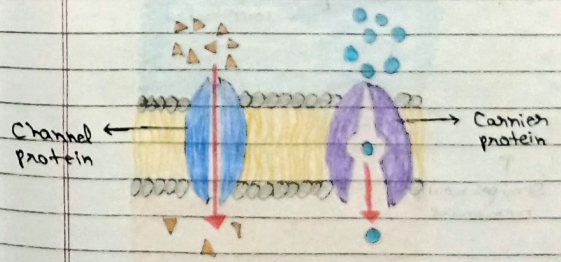
- channels
- Carriers

Examples

- Glucose transport through certain GLUT transporters
- Movement of some ions through ion channels

Characteristics

- Passive process
- No direct ATP requirement
- Occurs down the gradient
- Requires a specific membrane protein
- Can show saturation when transport proteins become fully occupie



Passive transport : Facilitated diffusion

1.c. Osmosis

Osmosis is the movement of water across a selectively permeable membrane from a region of relatively higher solute concentration

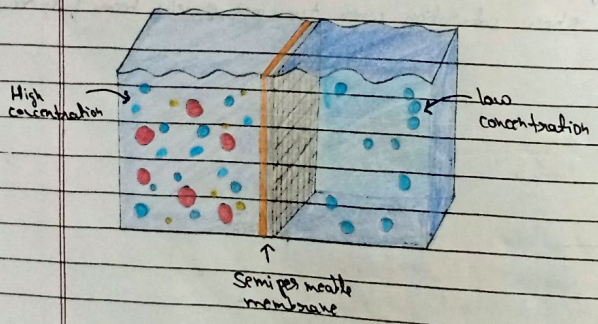
- Water moves until the effective osmotic different is reduced.

- In biological membranes, water movement occurs partly through specialized water channels called aquaporins.

Importance

Osmosis is important in maintaining:

- Cell volume
- Water balance
- Distribution of body fluids



Osmosis

2.a. Primary active transport

Primary active transport uses energy directly from ATP hydrolysis to transport substances against their gradient

- The transport proteins involved are called pumps.

Sodium - Potassium Pump

- The most important example is the Na^+/K^+ ATPase.
- For each ATP molecule used, it generally transports:

3 Na^+ out of the cell
2 K^+ into the cell

Function

- Maintain high intracellular K^+ concentration
- Maintains low intracellular Na^+ concentration
- Contributes to resting membrane potential
- Helps regulate cell volume
- Provides the Na^+ gradient used for secondary active transp

2.b. Secondary Active transport

Secondary active transport does not use ATP directly at the transport protein.

Instead, it uses the stored energy in an ion concentration gradient, commonly the Na^+ gradient, which has been established

by primary active transport

It occurs in two forms:

Symport: Two substances move in the same direction.
Example: Sodium-glucose cotransport in the kidney

Antiport: Two substances move in opposite directions
Example: Sodium-calcium exchange.

3.a. Endocytosis

Endocytosis is the process by which a cell takes substances into the cell by forming a vesicle from the plasma membrane.

Major types include:

Pinocytosis: Uptake of extracellular fluid and dissolved substances

Phagocytosis: Uptake of large particles or microorganisms by specialized cells. Ex- Macrophages engulfing microorganisms.

Receptor-mediated endocytosis: Specific molecules bind to receptors on the cell surface and are then taken into the cell.

Ex- Uptake of LDL particles.

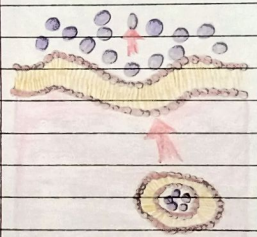
3.b. Exocytosis.

Exocytosis is the process by which substances are released from the cell through fusion of intracellular vesicles with the plasma membrane.

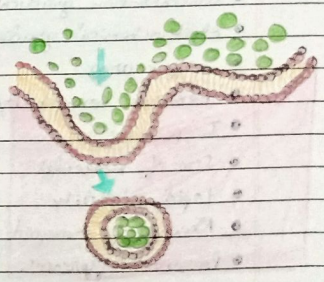
It is important for secretion of:

- Hormones
- Neurotransmitters
- Digestive enzymes
- Other cellular products

Example: A secretory vesicle fuses with the plasma membrane and releases its contents into the extracellular space.



Exocytosis



Endocytosis

Filtration

Filtration is the movement of water and small dissolved

substances across a membrane due to a pressure difference.

- It is particularly important in the formation of urine in the kidneys.

- In the renal glomerulus, blood pressure promotes filtration of water and small solutes from the blood into the renal capsule.

Factors Affecting Membrane Transport

The rate of and direction of transport affected by:

- Concentration gradient
- Electrochemical gradient
- Surface area
- Membrane Permeability
- Temperature
- Size of molecules
- Lipid solubility
- Number of transport proteins
- Pressure differences.

HOMEOSTASIS

Introduction

The human body is continuously exposed to changes in the internal and external environment. Despite these changes, the body environment maintains a relatively stable required for normal functioning of cells and tissues.

- The maintenance of a relatively constant internal environment is called homeostasis.

- The term homeostasis was introduced by Walter Cannon.

- Homeostasis does not mean that all conditions remain absolutely constant. Instead, physiological variables are continuously regulated within a normal range.

Examples of variables regulated by the body include:

- Body temperature
- Blood glucose concentration
- Blood pressure
- Blood pH
- Osmolality
- Calcium concentration
- Oxygen and carbondioxide levels.

Internal Environment

The cells of the body are surrounded by extracellular fluid (ECF).

The extracellular fluid provides the immediate environment in which cells live.

It supplies cells with:

- Oxygen
- Nutrients
- Water
- Electrolytes

It also receives:

- Carbon dioxide
- Metabolic waste products

Maintenance of a suitable extracellular environment is therefore essential for normal cellular function.

Components of a homeostatic control system

A basic homeostatic mechanism has three main components:

1. Receptor: The receptor detects a change in a regulated variable.

2. Control center: The control centre receives information from the receptor and determines the appropriate response.

3. Effector: The effector produces a response that helps restore the variable towards its normal range.

Basic sequence	Example
Stimulus	Body temperature ↑
↓	↓
Receptor	Thermoreceptors
↓	↓
Control centre	Hypothalamus
↓	↓
Effector	Sweat glands + skin blood vessels
↓	↓
Response	Increased heat loss
	↓
	Body temperature - Normal

Negative Feedback

Negative feedback is the most important mechanism used in homeostasis. In negative feedback, the response produced by the control system opposes the original change.

Example: Regulation of body temperature

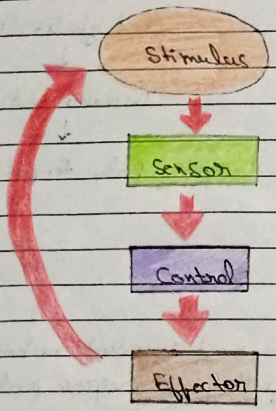
When body temperature rises, the hypothalamus activates mechanisms such as:

- Sweating
- Vasodilation of skin blood vessels

These increase heat loss and bring body temperature towards the normal range.

Other examples

- Blood glucose regulation
- Blood pressure regulation
- Regulation of blood calcium
- Regulation of body fluid balance



Positive feedback

In positive feedback, the response increases or reiforces the original stimulus.

Unlike negative feedback, it does not oppose the initial change.

Positive feedback is less common in normal physiology and usually continues until a specific event is completed.

Example:

Childbirth cervical stretching
↓
Oxytocin release
↓
Uterine contraction
↓
Further cervical stretching
↓
Further oxytocin release

This continues until the delivery of baby.

Blood clotting: Activation of clotting factors promotes activation of additional clotting factors, helping the clotting process proceed rapidly until bleeding is controlled.

Regulation of Body temperature

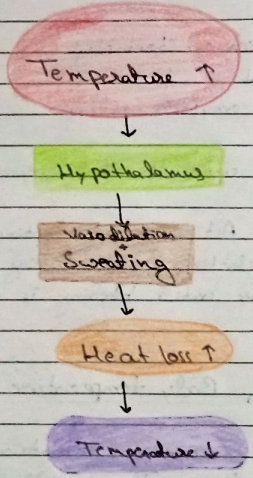
Body temperature must be maintained within a relatively narrow cellular activity. The hypothalamus acts as an important control centre for thermoregulation.

When body temperature increases:-
The body increases heat loss through:

- Sweating
- Vasodilation of skin blood vessels.

When body temperature decreases :-
The body reduces heat loss and increases heat production

- Vasoconstriction
- Shivering
- Increased metabolic heat production.

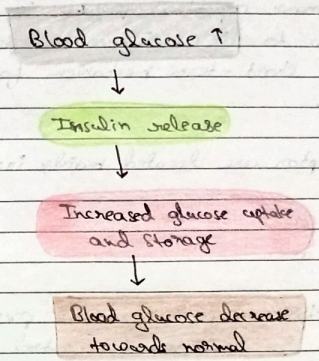


Regulation of Blood Glucose

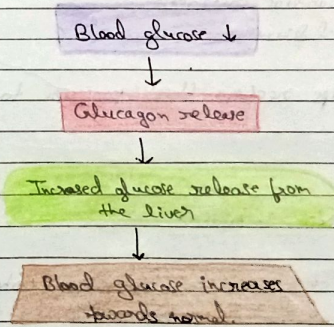
Blood glucose is maintained within a controlled range mainly by the hormones insulin and glucagon.

When blood glucose increases :-

After a meal:



When blood glucose decreases :-



Regulation of Blood pressure

Maintenance of blood pressure is essential for adequate blood flow to tissues. The baroreceptor reflex is an important short-term mechanism for blood pressure regulation.

Baroreceptor are located mainly in:

- Carotid sinus
- Aortic arch

They detect changes in arterial pressure and send information to cardiovascular control centres in the brain.

The autonomic nervous system then changes:

- Heart rate
- Force of cardiac contraction
- Blood vessel Diameter

This helps restore blood pressure towards the normal range.

Homeostatic imbalance

Homeostasis may be disturbed when the body's regulatory mechanisms are unable to maintain normal conditions.

Examples include:

Diabetes mellitus: Abnormal regulation of blood glucose results in persistent hyperglycaemia.

Dehydration: Excessive loss of water disturbs fluid and electrolyte balance.

Hypertension: Persistent elevation of blood pressure represents an abnormal state of cardiovascular regulation.

Fever: During fever, the regulated body temperature is temporarily set at a higher level, commonly in response to infection or inflammation.

Homeostatic imbalance can contribute to disease when normal regulatory mechanisms are disturbed or overwhelmed.

Genesis of resting membrane Potential

Introduction

Resting membrane Potential (RMP) is the electrical potential differ across the plasma membrane of a cell when it is at rest

⇒ The inside of the cell is negative relative to the outside. In neurons, the normal resting membrane potential is approximately -70

⇒ RMP is essential for the excitability of nerve and muscle cells and plays an important role in the generation of action potentials.

Definition

Resting Membrane Potential (RMP) is the electrical potential differen the inner and outer surfaces of the cell membrane when the cell i its resting state.

The inside of the cell is negative relative to the outside.

Normal RMP of neuron : -70 mV.

3. Ionic Distribution

Ion	Inside	Outside
Na^+	low	High
K^+	High	low
Ca^{2+}	Very low	High
Cl^-	Low	High
Protein	High	Low

4. Factors Responsible for Genesis of RMP

A. Selective Permeability of Cell Membrane

⇒ The resting membrane is more permeable to potassium than sodium due to the presence of potassium leak channels.

⇒ Potassium ions diffuse outside the cell along their concentration gradient. This loss of positive charge makes the inside of the cell negative.

⇒ This is the most important factor responsible for RMP.

B. Diffusion of Sodium Ions

⇒ A small amount of sodium enters the cell through sodium leak channels.
⇒ This inward movement of positive charges makes the inside less negative. However, sodium permeability is much lower than potassium permeability at rest.

C. Intracellular Negatively charged proteins

⇒ The intracellular fluid contains negatively charged proteins and other organic anions that cannot freely cross the cell membrane.

⇒ These negative charges contribute to the negativity of the intracellular fluid.

D. Sodium-Potassium Pump

⇒ The Na^+/K^+ ATPase activity transports ions using energy from ATP.

For every cycle:
- 3 Na^+ ions move outside
- 2 K^+ ions move inside
- 1 ATP molecule is consumed.

The pump maintain the ionic concentration gradients and contribute intracellular negativity.

5. Mechanism of Genesis of RMP

1. Potassium concentration is high inside the cell, while sodium concentration high outside
2. The resting membrane is highly permeable to potassium
3. Potassium diffuses outside through leak channel
4. loss of positive charges makes the inside negative
5. The negative electrical force attracts potassium back into the cell.
6. electrical and chemical forces attracts potassium back into the cell
7. The sodium-potassium pump maintains the ionic gradients.

Thus, a stable resting membrane potential is established.

6. Equilibrium Potential

Equilibrium potential is the membrane potential at which the electrical force balances the chemical force acting on an ion

→ The potassium equilibrium potential is approximately −90 mV.

→ Since the resting membrane is mainly permeable to potassium, RMP is close to its equilibrium potential.

However, neuronal RMP is approximately −70 mV because the membrane also have some permeability to sodium and other ions.

7. Nernst Equation

The nernst equation calculates the equilibrium potential of a single ion

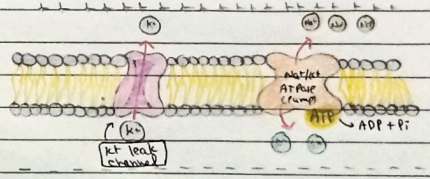
At 37°C: $E = 61.5 \log_{10} (\text{outside conc.} / \text{inside conc.})$ mV

For potassium: $E_K = 61.5 \log_{10} ([K^+] \text{outside} / [K^+] \text{inside})$

8. Goldman-Hodgkin-Katz equation

The goldman-Hodgkin-katz equation calculates membrane potential by considering the concentration gradients and relative permeability of different ions.

Since potassium permeability is highest at rest, potassium has the greatest influence on RMP.



10. Significance

1. Maintains nerve and muscle excitability.
2. Helps generate action potentials.
3. Maintains ionic gradient
4. Essential for muscle contraction.
5. Supports normal functioning of excitable tissues.